École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

Created & Coordinated by: M. Salman Nadeem

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INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check **Zeus** platform for the date & time)	1	**Primer, TCP/IP stack, TCP/IP stack operations, Class exercises**	**3**
(check **Zeus** platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check **Zeus** platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check **Zeus** platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check **Zeus** platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check **Zeus** platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check **Zeus** platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check **Zeus** platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
		Total	***24 hours***
(check **Zeus** platform for the date & time)	EXAM		1

GRADING criteria:

Class attendance and responsiveness: 10%
Class assignments (practical work): 40%
Final evaluation (Quiz & Exam): 50%

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Notes & Collaboration

- MS Teams Channel:
  'Introduction to Network protocols and architecture'
  Course specific channel to collaborate
  - Will be used to:
    - Publish course related announcements
    - Provide course slides/material
    - Receive assignments and projects
- Course Mindmap:
  - For better organization and easy refreshing of course topics
  - Access link (read-only):
    https://www.mindomo.com/mindmap/d6ee466294704f0890103638e18b50f2

→ It is possible to forward questions in advance regarding INPA course topics you would like to see discussed (to the extent possible and in line with the scope of the course).

Check policies in the course outline

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Lecture 1 Outline

- **Primer**
  - **Brief overview of Connection setup**
  - **Class exercise 1**
- TCP/IP Stack
  - Overview: Network protocols, types, topologies & architecture
  - Class exercise 2
  - TCP/IP & OSI Model
- TCP/IP Stack
  - Data transfer example
  - TCP/IP Layer operations
  - Class exercise 3

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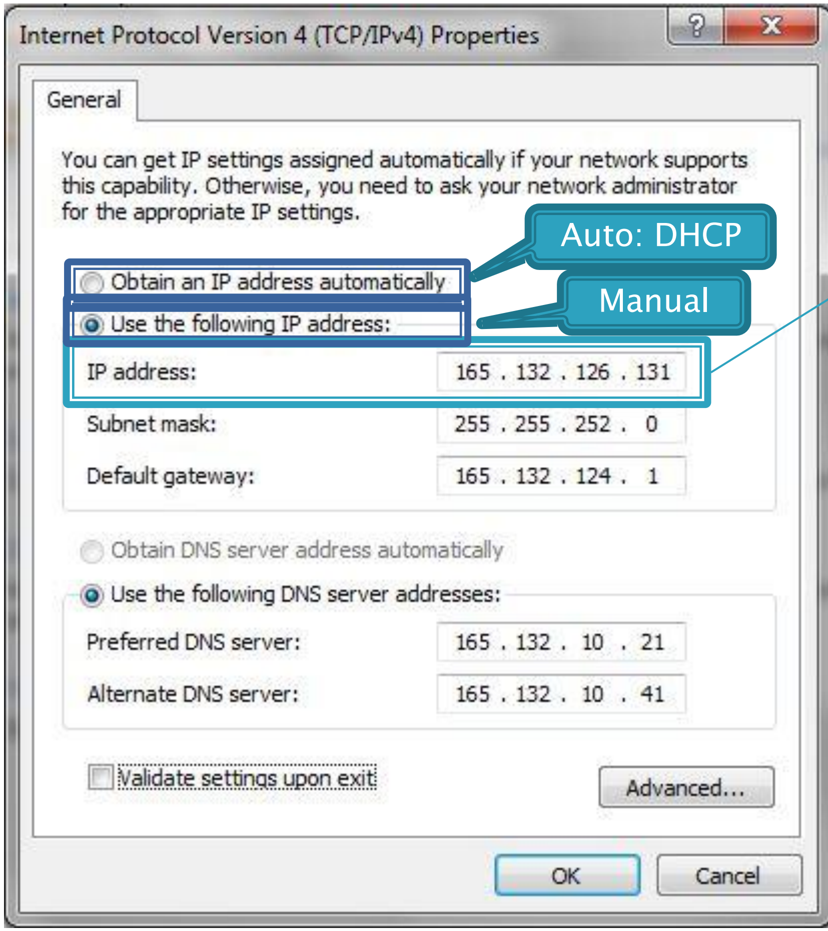
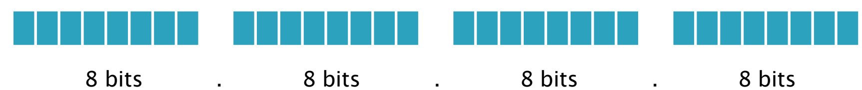
IPv4 Network connection setup



Screenshot from: IPv4 connection properties dialog captured from locally installed Windows OS

IP address is assigned to an interface port

- Device Internet interface: 165.132.126.131
- IPv4: **32 bits** (4 bytes)/4 Octets address format



8 bits: 00000000 ~ 11111111 (binary)
= 0 ~ 255 (decimal)

8bits = 1 Byte = 1 Octet

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Binary & Decimal values

165.132.126.131
(Dotted decimal notation)

10100101.10000100.01111110.10000011
(Dotted binary notation)

- Equivalent decimal value?

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	Formula: 2^n
1	1	1	1	1	1	1	1	Binary form
128 +	64 +	32 +	16 +	8 +	4 +	2 +	1 = 255	Decimal form

The largest decimal number that can be stored in an IP address octet is **255** (the significance of this should become evident later in this course).

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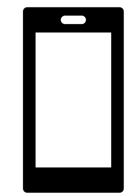
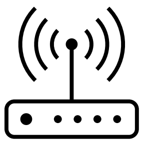
IP Address Assignments

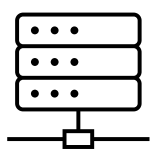
- Devices may have multiple Interfaces, and each can have its own IP address e.g. smartphone:
  - Mobile communication standards
    - 2G – Global System for Mobile (GSM)
    - 3G – Universal Mobile Telecommunications System (UMTS)
    - 4G – Long Term Evolution (LTE)
    - 5G – New Radio (NR), …
  - Wi-Fi -> Specific IP address
    - IEEE 802.11 a/b/g/b/ac (2.4, 5 GHz)

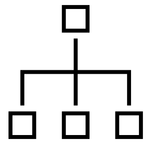


Other examples?

- Server e.g., having multiple NICs
- VLAN (e.g., sub-interface)
- Perimeter and DMZ isolation
- …







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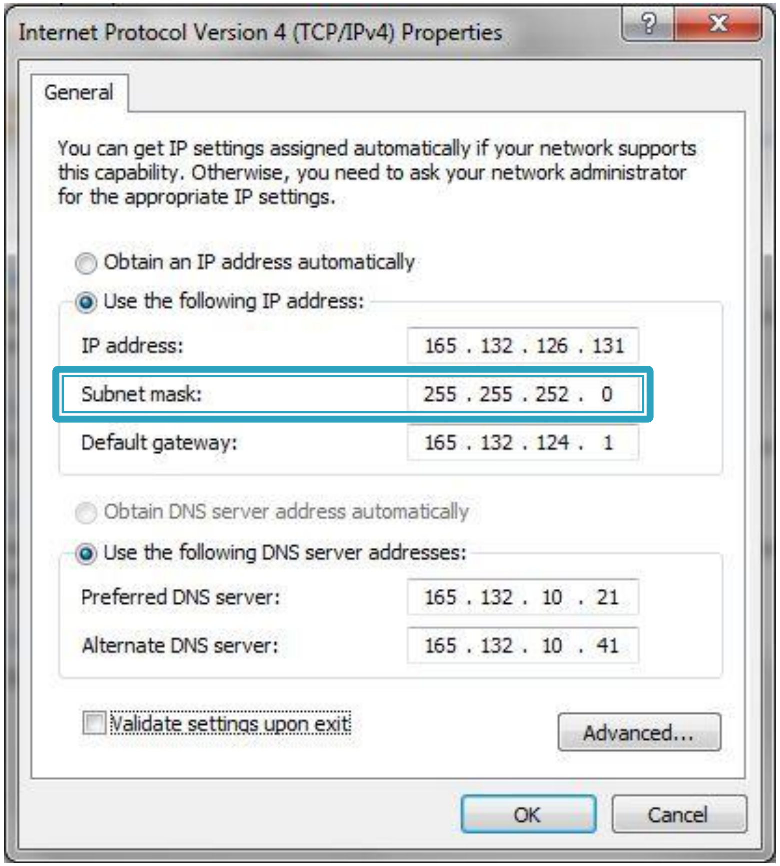
Subnet Mask

- Internet is divided into Subnets, and Subnets are divided into smaller Subnets…
  - Subnet Mask is based on the size of the Subnet that the Client (device) is connected to
- Used to mask (filter) the IP address (in IP packets) to determine the respective subnet
  - 255.255.252.0 =1111111.11111111.11111100.00000000
- Subnet Mask can be used to find the Subnet size e.g.
  - 2^{10} = 1,024 IP addresses are on this subnet



Screenshot from: IPv4 connection properties dialog captured from locally installed Windows OS

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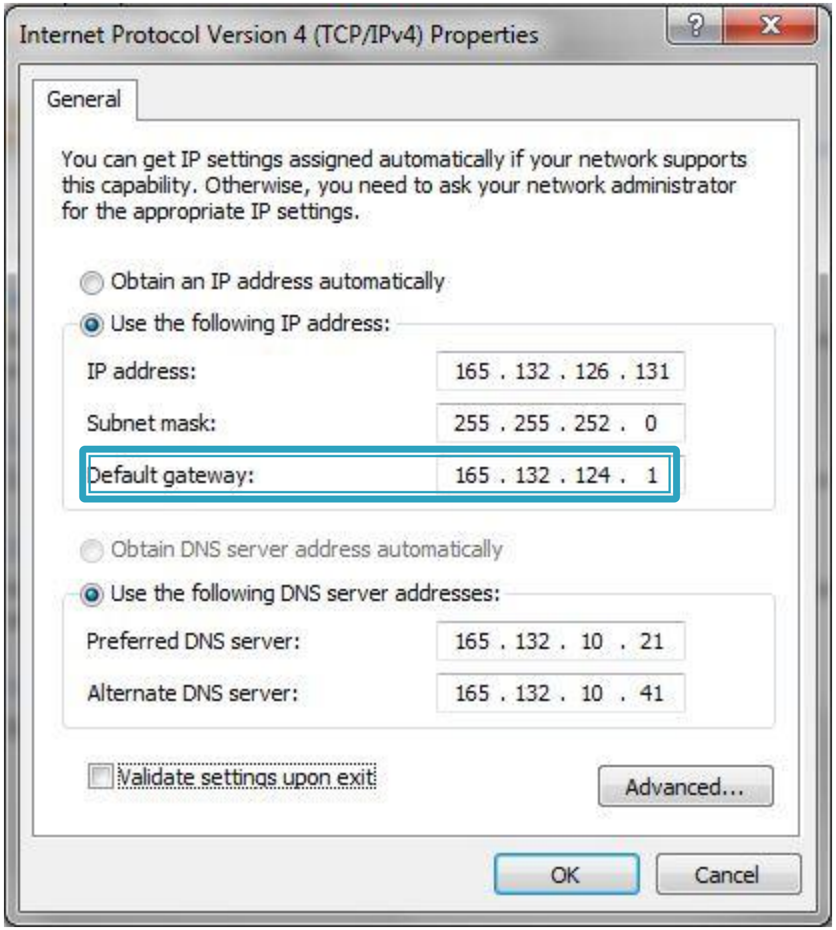
- Default Gateway:
  - The dedicated Internet Router that will send and receive all Internet IP packets for this device
  - Device will access the Internet (i.e., send and receive all IP packets) through this Gateway



Screenshot from: IPv4 connection properties dialog captured from locally installed Windows OS

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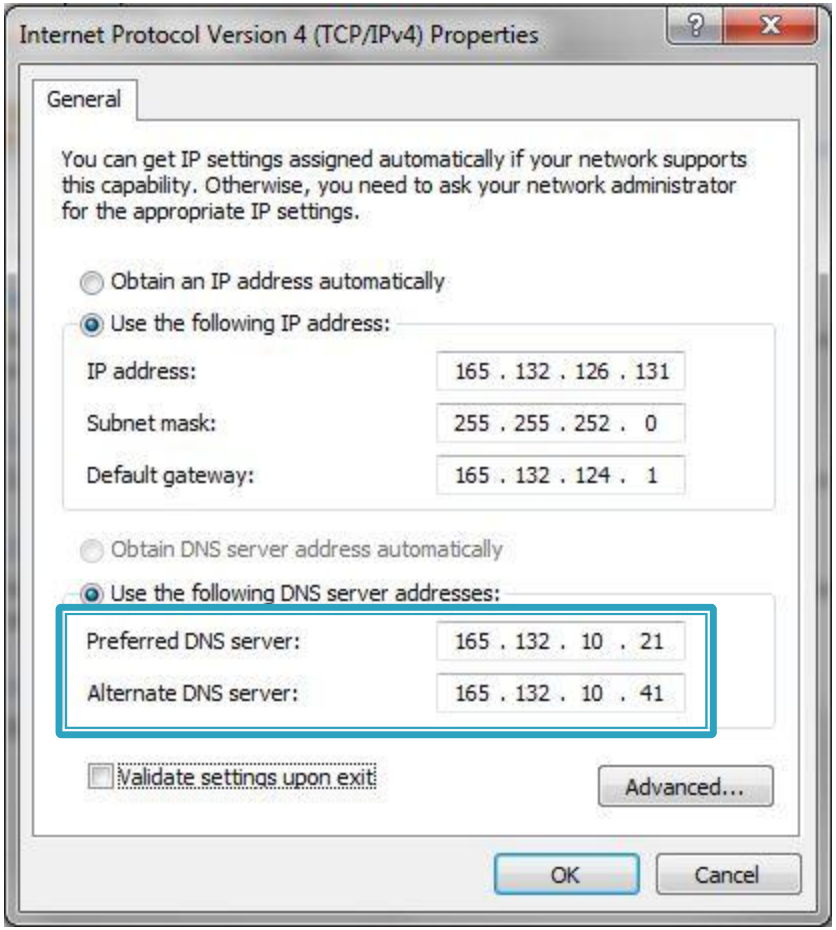
- DNS (Domain Name Server)
  - DNS is a server that converts hostnames in to IP addresses
  - 'Preferred' is the main one, 'Alternate' one is the backup
  - Hostname examples
    - E-mail address: ???@epita.fr
    - Domain name: www.epita.fr



Screenshot from: IPv4 connection properties dialog captured from locally installed Windows OS

- **Can we change it?**
  - **Why?**
  - **How?**

For more info: https://dnsprivacy.org/wiki/

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Physical setup

- How do we actually connect to the internet?

Feature	Method				
	Baseband	Broadband			
	Dial-up	DSL	Cable	Satellite	Cellular (3G / 4G / 5G)
Equipment	Modem	DSL modem, Ethernet card	DSL modem, Ethernet card	Satellite dish, modem, receiver (e.g., Starlink Kit)	Smartphone, mobile hotspot, or cellular modem
Requires	Phone service and ISP	Service and local or short distance to ISP office	Cable connection	Clear view of the sky, satellite service provider (e.g., Starlink)	Cellular coverage and SIM card / data plan
Connection type	Dial to connect	Always on, dedicated	Always on, shared	Always on [Wireless satellite link (via Low Earth Orbit satellites)]	Always on, wireless
Typical speed	56 Kbps (max)	Varies by DSL type (e.g., ADSL, VDSL), typically between 128 Kbps and 100 Mbps.	Typically 10 Mbps to 1 Gbps, depending on provider and network congestion	25 Mbps – 220 Mbps (download), 5 Mbps – 20 Mbps (upload)	3G: ~1-10 Mbps 4G LTE: ~10-100 Mbps 5G: 100 Mbps to 1+ Gbps

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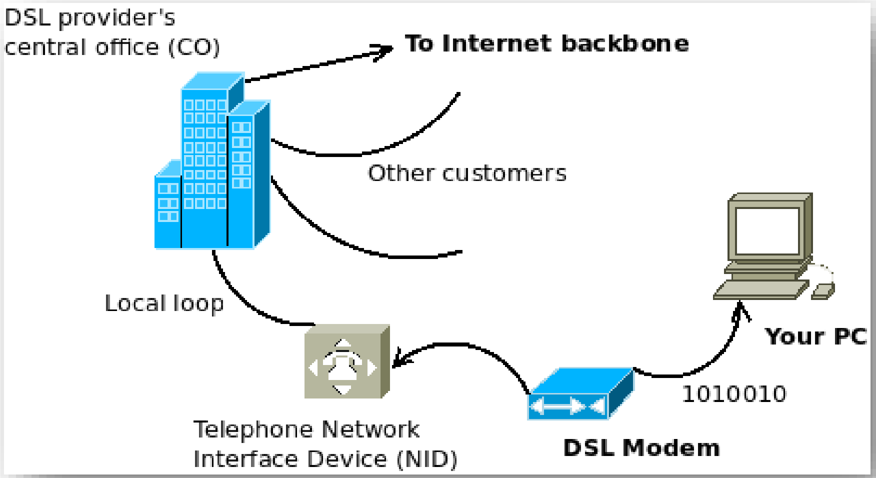
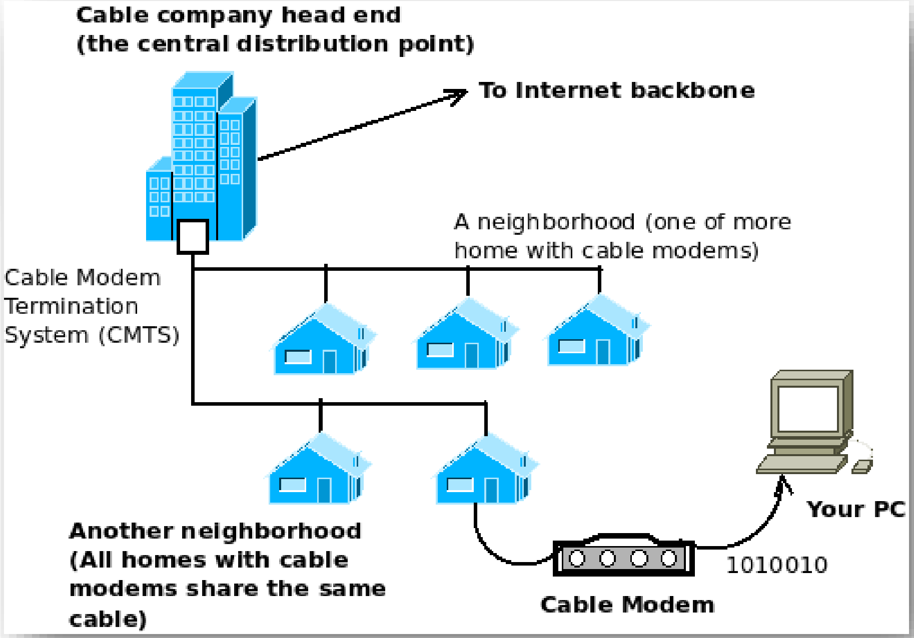
Physical setup (DSL & Cable)

DSL



Cable modem



That seems nice, but will it protect your connection from intruders ❓

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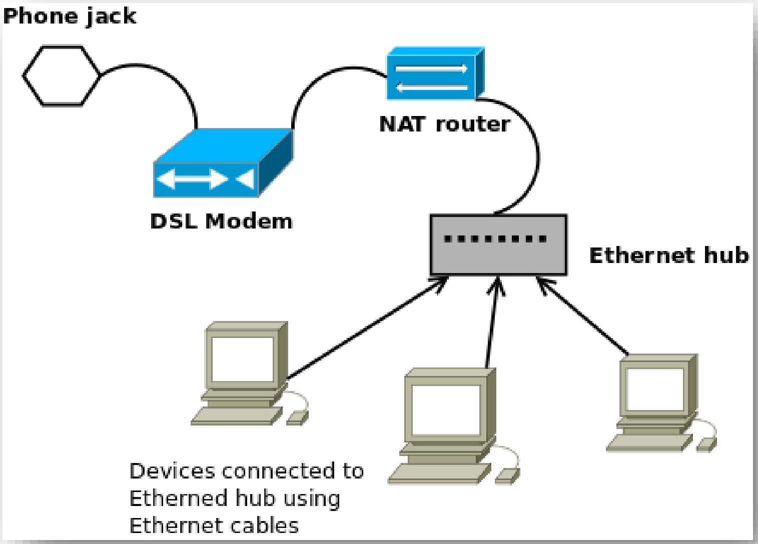
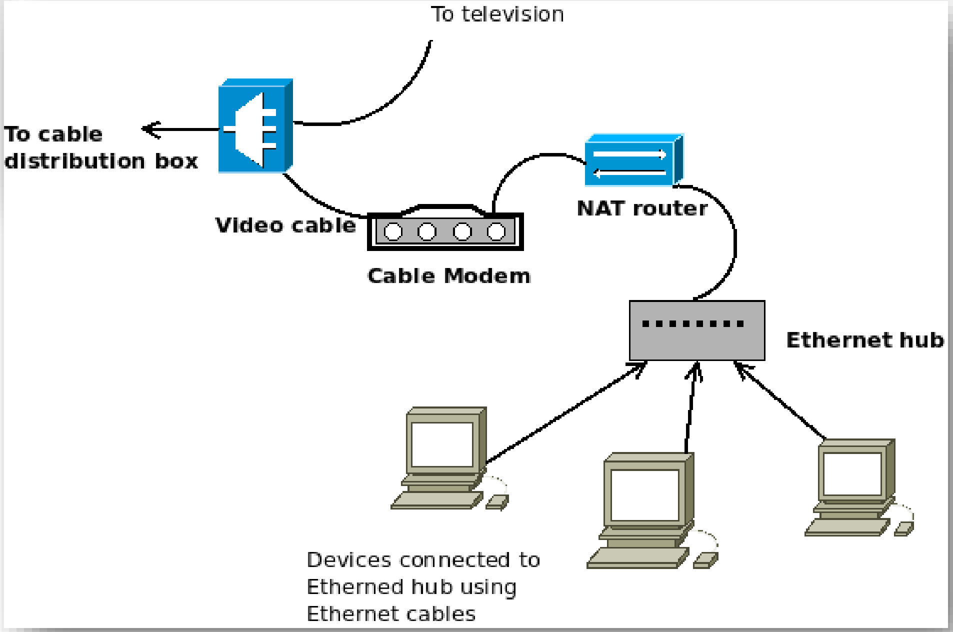
Physical setup (NATed)

DSL (with NAT)



Cable modem (with NAT)



ℹ NAT: Will hide your internal IP address(es) from outside networks

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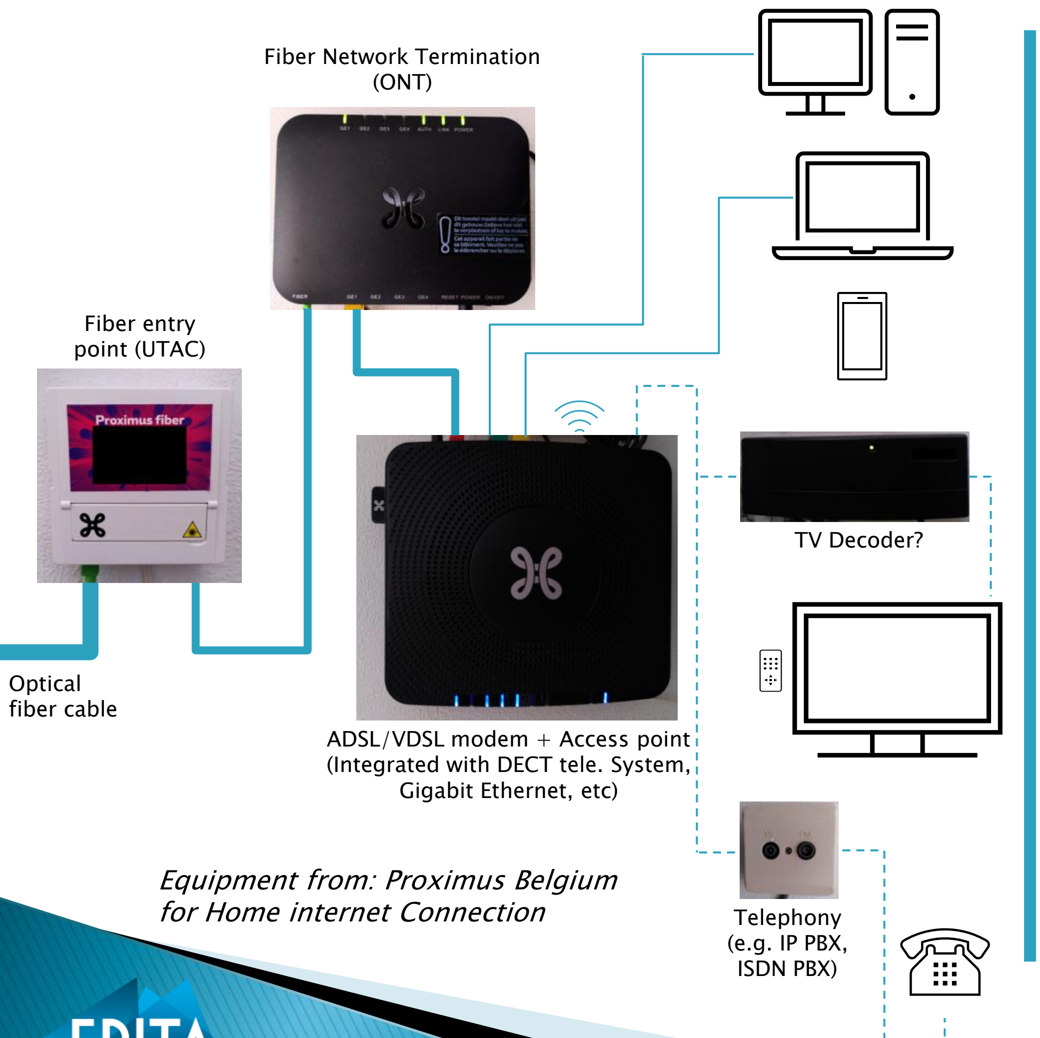
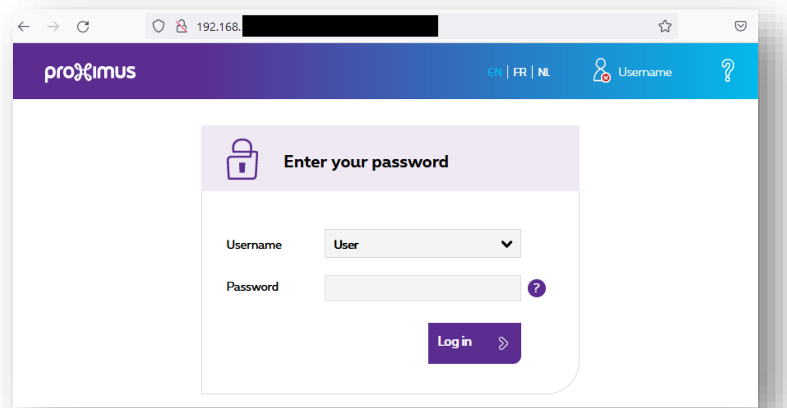
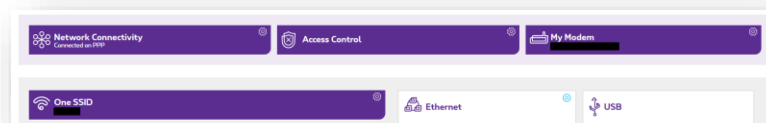
Connection (physical) setup (client side) – Fiber cable connection



Equipment from: Proximus Belgium for Home internet Connection



ADSL/VDSL modem + Access point (local) Interface



ADSL/VDSL modem + Access point (local) Interface Dashboard

Screenshots from: Proximus Belgium router web interface (local network login screen) captured locally

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Exercise 1: Practical work

- Answer following questions:
  1. Highlight each parameter of your TCP/IP connection configuration (IPv4)?
  2. Does your device has multiple IP assignments? (if yes, then highlight them)
  3. DNS server:
     - What is your DNS server?
     - Why is it preferable to protect DNS traffic?
     - Change your DNS server to other DNS server of your choice? (for the purpose of this exercise)
  4. Which connection (physical) setup (e.g., DSL, Cable model, …) are you directly connected to? **(visualize your answer)**

     → Last slide shows an example

     → OR use a network simulator e.g. Cisco Packet Tracer (direct link)

     Others: https://nil.uniza.sk/network-simulation-virtualization-software-list

     - Would you like to switch to a different connection setup? If yes then, why?

Deadline: See 'Teams' Assignment section

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Lecture 1 Outline

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- **TCP/IP Stack**
  - **Overview: Network protocols, types, topologies & architecture**
  - **Class exercise 2**
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  - Data transfer example
  - TCP/IP Layer operations
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Network Protocols

- Internet: A global system of interconnected computer networks
  - Network protocols can be completely different in nature
  - TCP/IP is what links them
- Protocols allow networks to exchange data
  - Prescribed guide for conduct or action/or series of actions
  - Devices know in advance about the data to be exchanged along with its precise format
- They are standards that specifies:
  - The format of the messages
  - How to handle errors

? Who defines protocol as a standard?

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Network types

- There are numerous types of Networks (as per their **size** and **reach**)
  - LAN (Local Area Network): Close devices/nodes are connected using Ethernet cables/wireless media (e.g., Home network, same building network, ...); 0-2 km
  - MAN (Metropolitan Area Network): It is bigger than LAN e.g., usually spans all over the city using copper wires/high speed fiber-optics, …; 2-50 km
  - WAN (Wide Area Network): It is made up of multiple MAN's and should span over 30 miles by convention/definition. It uses satellites, fiber-optics, copper wires etc; 50+ km
  - VPN (Virtual Private Network): Private networks which contains clients (using VPN client application) from different locations under a Virtual Network
  - …

What network type is the Internet?

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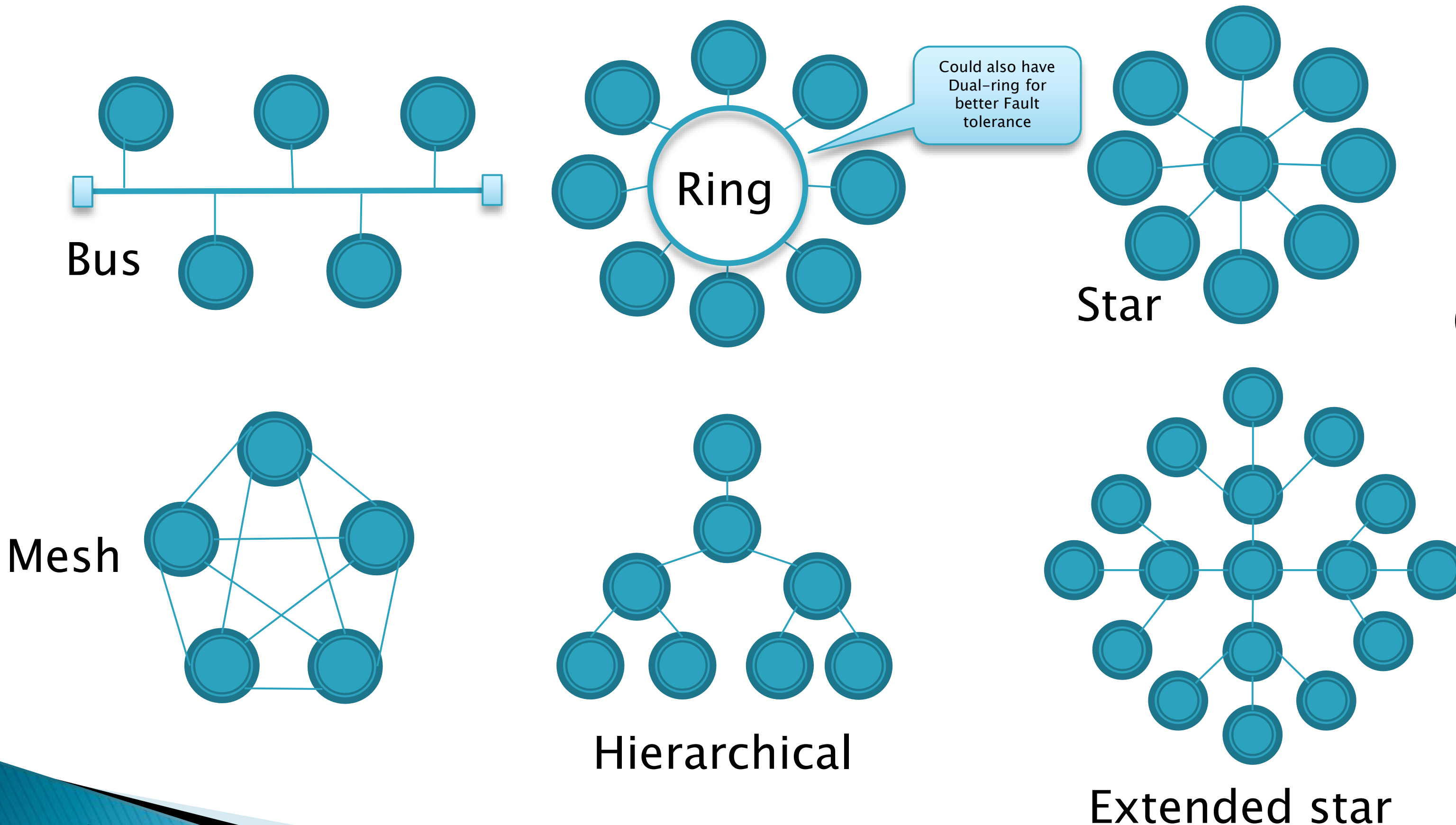
Network topologies

- Every network type has a topology (physical layout) for each of its components or for the system as a whole, with key decision factors including **cost**, **fault tolerance**, and **speed**



What is the network topology of the Internet?

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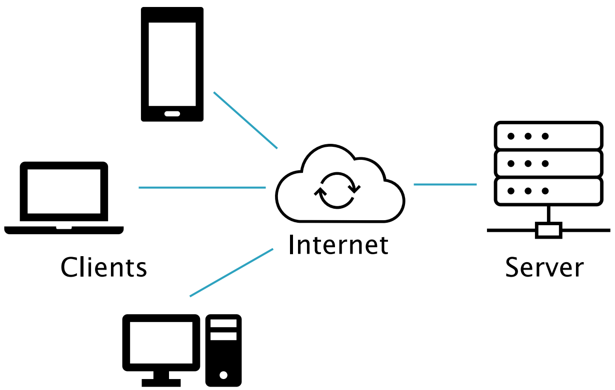
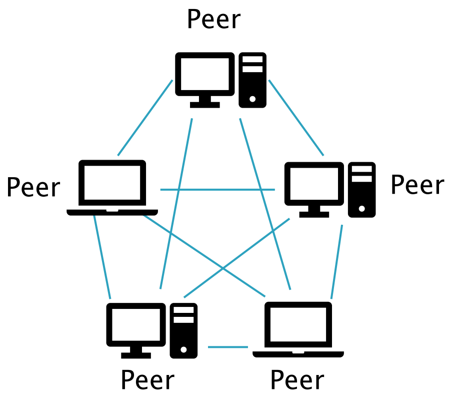
Network architecture

- Each of the Network or its part has a logical architecture and works accordingly
- Predominantly, we have:
  - Client/Server architecture (centralized)
  - Peer-to-peer architecture (de-centralized)
- Others: Mainframe-based networks (centralized), CDNs (distributed)





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Exercise 2: Practical work

- Based on your **existing** internet connection, answer following questions:
  1. Which protocol are you using to connect to the internet?
  2. What is the most probable network topology of your connection?
     - Make an informed guess!
  3. What is the logical network architecture of your connection? (in relation with your ISP)

Deadline: See ‘Teams’ Assignment section

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Layered network architecture

OSI 7 layer model	TCP/IP 5 layer model	Protocols	Protocol Data Unit (PDU)	Layer addressing
Application layer Presentation layer Session layer	Application layer	HTTP, FTP, SMTP, IMAP, DNS, Telnet, SNMP, …	Data	Data
Transport layer	Transport layer	TCP, UDP, …	Segment (TCP), Datagram (UDP)	Port address (Source, Destination)
Network layer	Internet layer	IP, …	Packets	IP address (Source, Destination)
Data link layer	Network access layer	Ethernet, Wi-fi, Bluetooth, 3G, LTE, …	Frames	MAC address (Source, Destination)
Physical layer	Physical layer	Copper, Fiber optic cables, Wireless transmitters	Bits	n/a

Open System Interconnection (OSI): ITU-T standard X.200 (ISO/IEC 7498-1)

DoD Model: Currently Maintained by IETF

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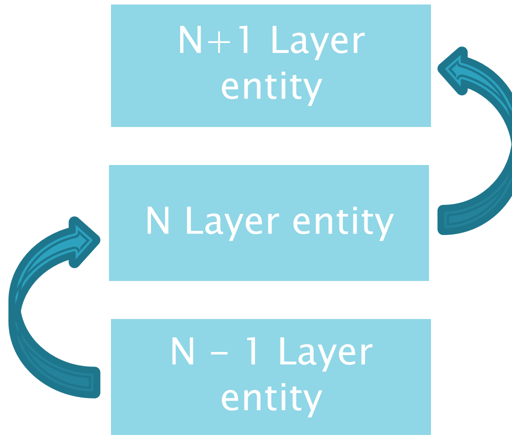
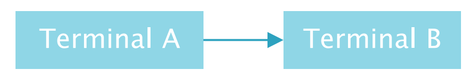
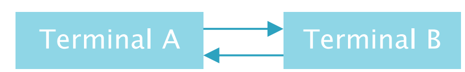
Layered network architecture (cont.)



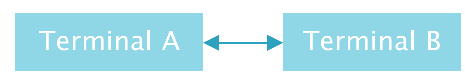
- Services are grouped in layered hierarchy
  - An entity of layer N uses only services of layer N-1
  - An entity of layer N provides services only to layer N+1
  - **TCP/IP model (however) does not require strict layering**
- Communication Transmission/Duplexity:
  - Simplex mode: Unidirectional e.g., Loudspeaker



  - Half-duplex mode: Either direction but only one-way at a time e.g., Walkie-talkie



  - Full-duplex mode: Both direction at same time e.g., Telephone



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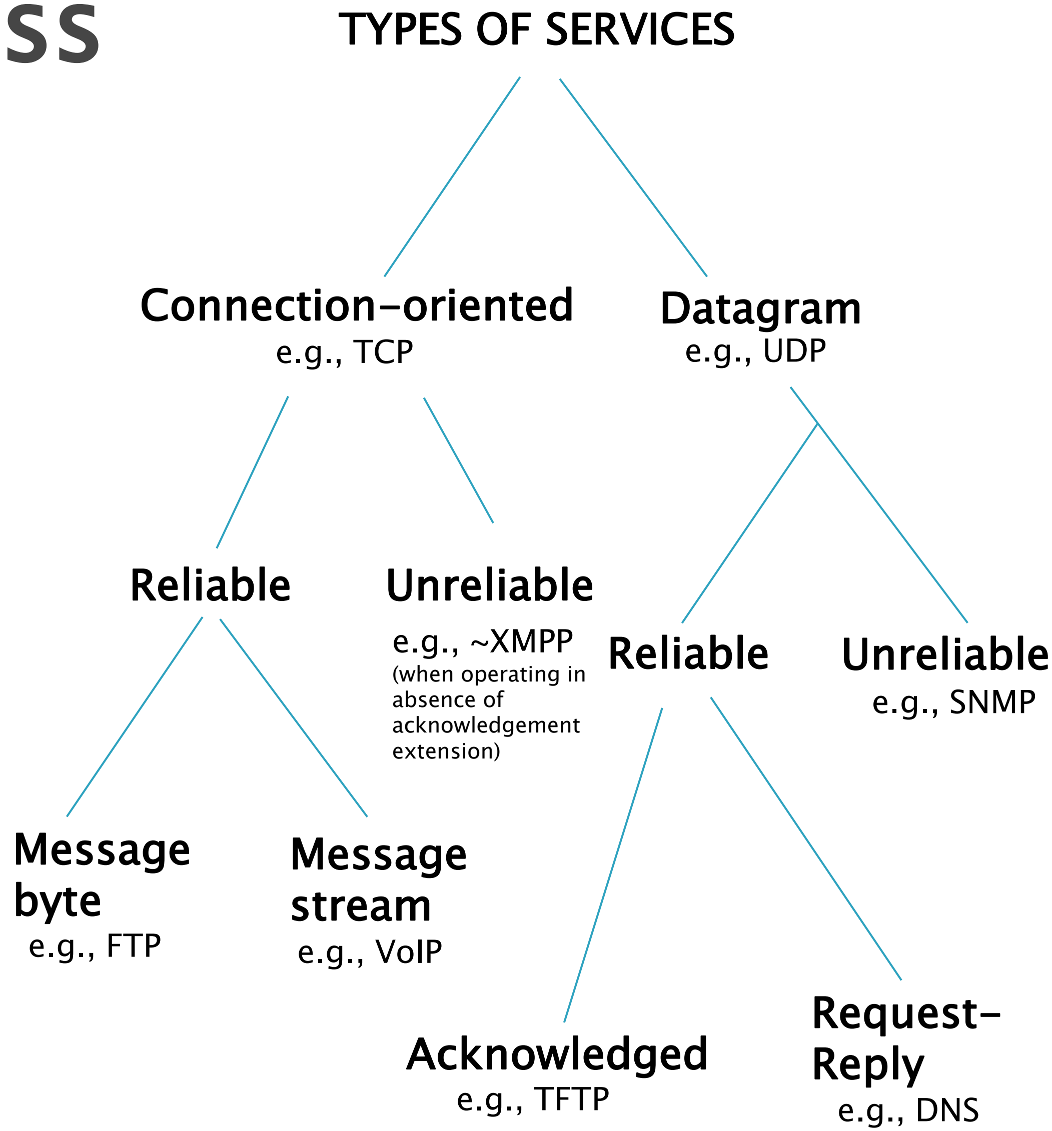
Connection-oriented vs connectionless

- Connection-oriented: *Telephone system*
  - Path setup before data is sent
  - Data do not need addressing. Circuit number is used
  - Virtual circuits: Multiple circuits on one wire
- Connectionless: *Postal system (also known as datagram)*
  - Complete address on each packet
  - The address decides the next hop at each routing point



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  - **Data transfer example**
  - **TCP/IP Layer operations**
  - **Class exercise 3**

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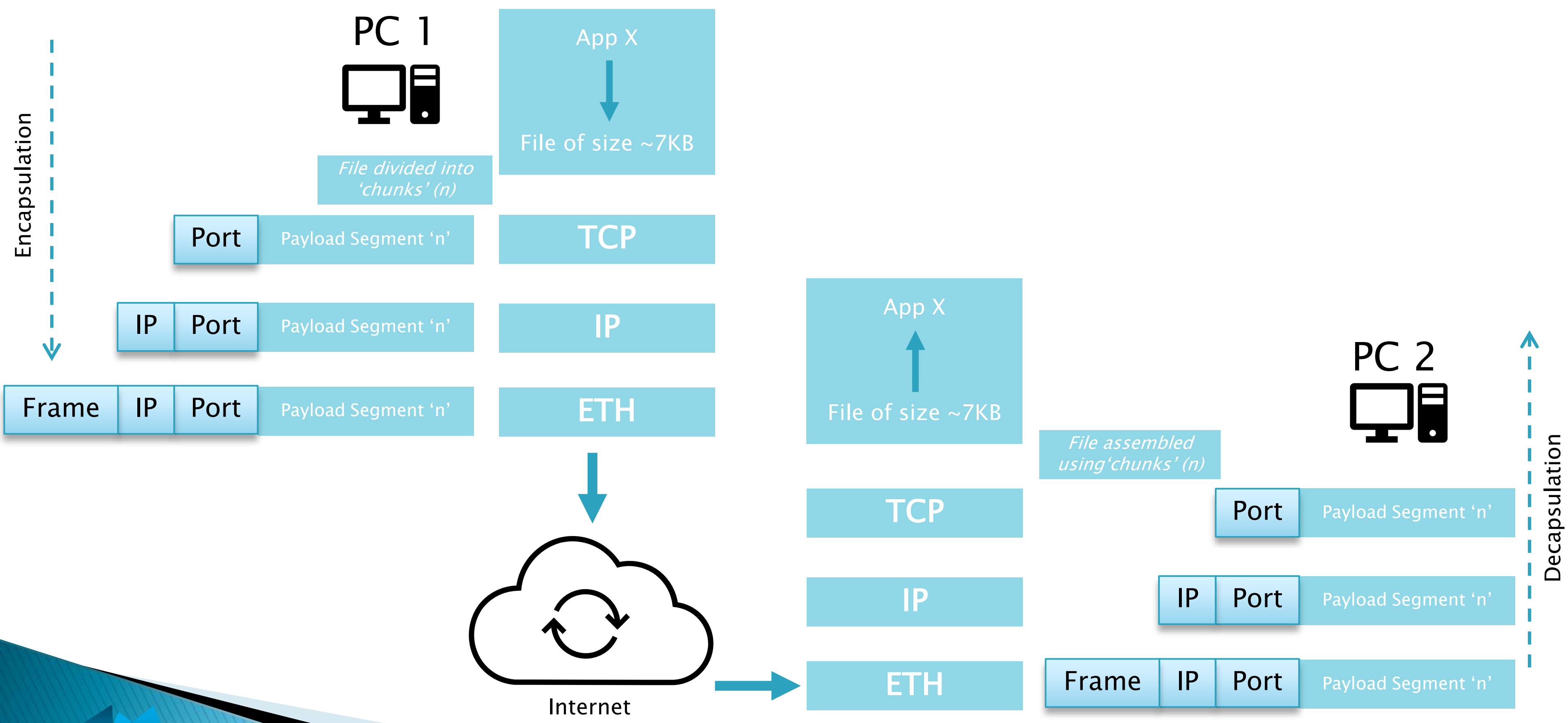
Data Transfer Example

Sending a 'file' from PC1 to PC2…



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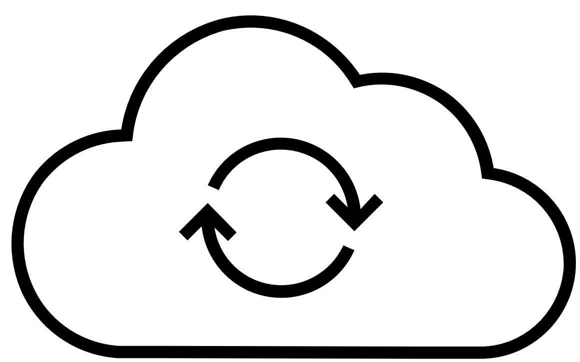
TCP/IP 5 layers & protocols

TCP/IP Layers

Layer 5: Application Layer
Layer 4: Transport layer
Layer 3: Internet layer
Layer 2: Network access layer
Layer 1: Physical layer

TCP/IP Protocols

E-mail
TCP
IP
Wi-Fi
2.4 Ghz



Internet

E-mail Server
TCP
IP
Ethernet
Cat 5

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Open Exercise: Cisco IOS (setup)

- Install Cisco Packet Tracer:

  Download URL: https://legacy.netacad.com/portal/resources/packet-tracer
  -> **To access all available Netacad and SkillsforAll services, please create a *Cisco Networking Academy account* using your EPITA Office365 email address.**

  Alternate link: https://mailfence.com/pub/docs/MSalman/web/Emulator-Simulator/

- Cisco IOS System
  - IOS (Internetwork Operating System) comes in NOS (Network Operating System) category
  - Works on router and cisco switch
  - Use to configure, the monitoring and to detect problems
  - Works with Specific CPU (Motorola 68030, Orion/R4600...)

You can use any other program of your choice:
https://nil.uniza.sk/network-simulation-virtualization-software-list

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Open Exercise: Cisco IOS (modes)

- Get yourself familiar:
  - Console port, using a physical access
  - Auxiliary port, for a connection through a modem
  - VTY (virtual TTY), access with Telnet or SSH
  - WEB
- Single mode (read only): hostname>
- Privilege mode (read,write): hostname #
- Configuration mode: hostname(config)#
- Special configuration mode:

  hostname(config-if)#

  hostname(config-router)#

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Open Exercise: Cisco IOS (workflow)

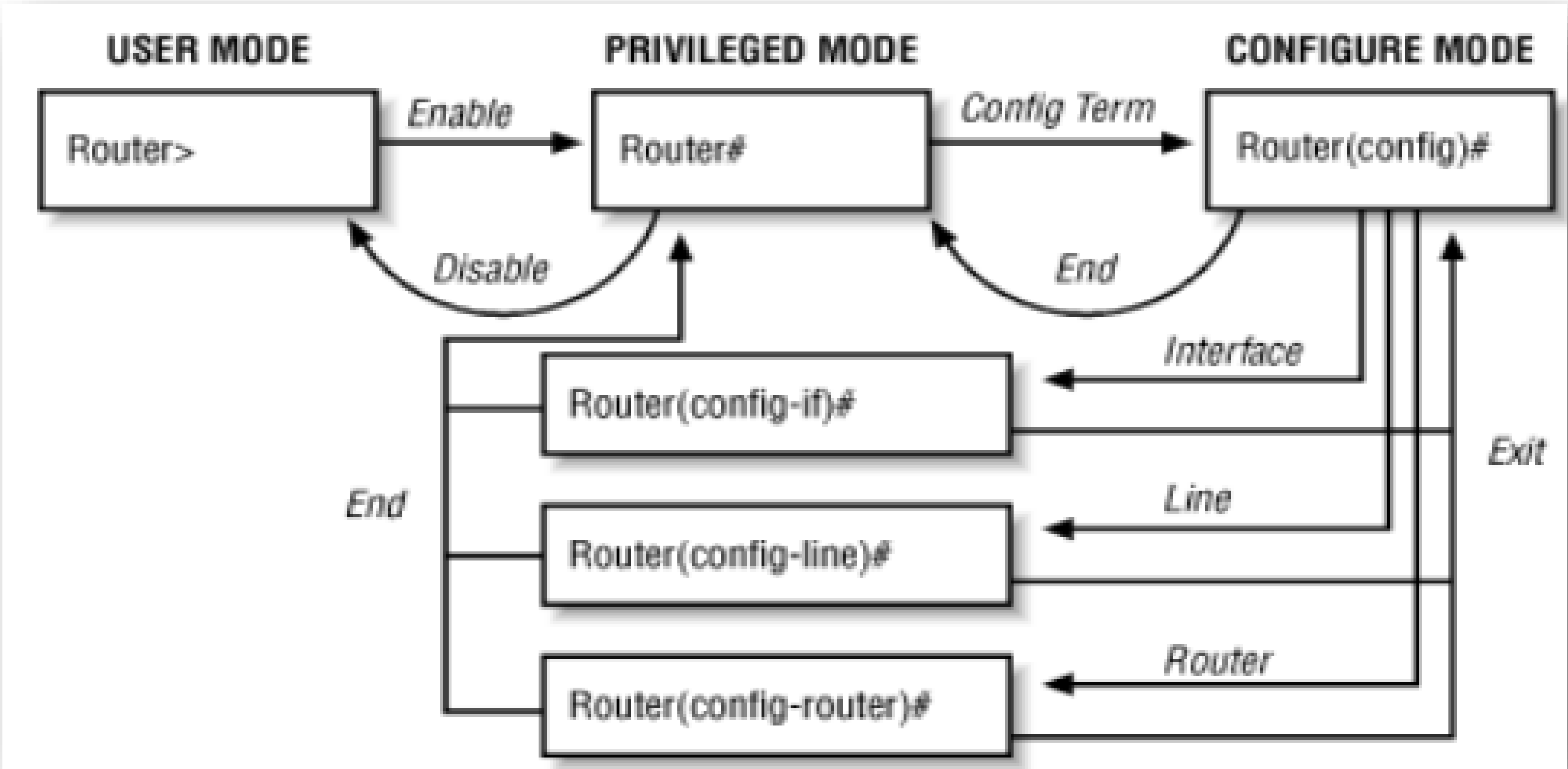

Diagram from: Cisco IOS in a Nutshell – User, Privileged, and Configuration Modes.
Source: https://www.oreilly.com/library/view/cisco-ios-in/156592942X/ch01s02.html

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Exercise 3: Practical work

- Use any FTP client to connect to FTP server of your choice (e.g., ftp.gnu.org) **OR** Use network simulator, and perform any (read/write) operation:
  1. List down the actions (in step-wise fashion with commands)
  2. Visualize/Map each connection with the respective TCP/IP Layer (e.g., take screenshots showing operations occurring at each of the TCP/IP layers and briefly explain them)

  -> Use network simulator e.g. Cisco Packet Tracer (direct link)

  Others: https://nil.uniza.sk/network-simulation-virtualization-software-list

Deadline: See 'Teams' Assignment section

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Lecture 1 ends here

- Course Slides: Go to MS Teams:
  Introduction to Network protocols and architecture
  → Files section

- Send your questions by email
  OR via direct message using MS Teams

- Thank You!

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